FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

CONCEPT RECAPITULATION TEST – IV

PAPER –1

TEST DATE: 26-04-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

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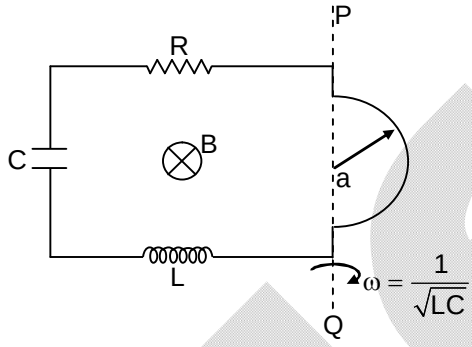
Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A wire shaped as a semicircle of radius a, is rotating about an axis PQ with a constant angular velocity $\omega = \frac{1}{\sqrt{LC}}$, with the help of an external agent. A uniform magnetic field B exists in space and is directed into the plane of the figure. (circuit part remains at rest) (left part is at rest)



(A) The rms value of current in the circuit is $\frac{\pi Ba^2}{R\sqrt{2LC}}$

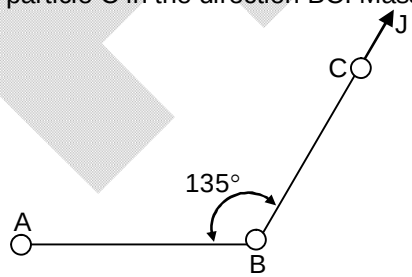
(B) The rms value of current in the circuit is $\frac{\pi Ba^2}{2R\sqrt{2LC}}$

(C) The maximum energy stored in the capacitor is $\frac{\pi^2 B^2 a^4}{8R^2 C}$

(D) The maximum power delivered by the external agent is $\frac{\pi^2 B^2 a^4}{4LCR}$

2. An electron revolves in first orbit in H atom, then:
(A) Current associated due to orbital motion of electron is 1.06 m A.
(B) Magnetic field at the centre of nucleus due to orbital motion of electron is 12.5 Tesla.
(C) First excitation energy of H atom is 10.2 eV.
(D) Current associated due to orbital motion of electron is 2.06 m A.

3. Three identical particles A, B and C lie on a smooth horizontal table. Light inextensible strings which are just taut connect AB and BC and $\angle ABC$ is 135°. An impulse J is imparted to the particle C in the direction BC. Mass of each particle is m. Choose the correct options.



(A) Speed of A just after the impulse imparted is $\frac{\sqrt{2}J}{7m}$

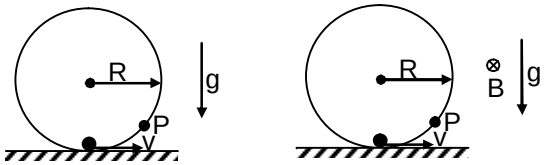
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(B) Speed of B just after the impulse imparted is $\frac{\sqrt{10}J}{7m}$

(C) Speed of C just after the impulse imparted is $\frac{3J}{7m}$

(D) Speed of A just after the impulse imparted is $\frac{2J}{7m}$

4. In case figure (a) and (b) a positive charge particle of mass m and charge q is given velocity 'v' at the lowest point of a smooth fixed sphere of radius R. In both cases circular motion is in the plane of the paper. In case (b) magnetic field is also switched on in inward direction. If at point P, v_1 and v_2, and v_2, N_1 and N_2 , are velocities and normal forces in two cases respectively, then



(a) $v = \sqrt{2gR}$ (b) $v = \sqrt{2gR}$

(A) $v_1 = v_2$ and $N_1 > N_2$ (B) $v_1 < v_2$ and $N_1 > N_2$

(C) $v_1 > v_2$ and $N_1 > N_2$ (D) $v_1 = v_2$ and $N_1 = N_2$

5. Three particle A, B, C each of mass m are moving on the same circular path and are always equidistant from each other. The radius of circle is R and the speed of each mass is same and equal to V. The only force acting on each particle is gravitational force due to other two particles. Then

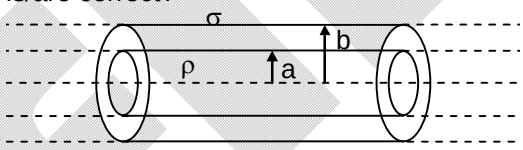
(A) Power of gravitational force by A on B is not equal zero.

(B) Power of gravitational force by A on B is zero.

(C) Power of the gravitational force on A is the frame of B is $\frac{\sqrt{3}mv^3}{2R}$.

(D) The ratio of magnitude of relative velocity of A with respect to B and A with respect to C is 1:1.

6. A long co-axial cable carries a uniform volume charge density ρ on inner cylinder and uniform surface charge density σ on outer cylinder. If radius of inner cylinder is a and radius of outer cylinder is b. It is found that this infinite cable is electrically neutral then which of the following is/are correct?



(A) $\frac{\sigma}{b^2} + \frac{\rho}{2a} = 0$

(B) $\frac{\sigma}{a^2} + \frac{\rho}{2b} = 0$

(C) Electricity field outside the cable increases linearly with radial distance.

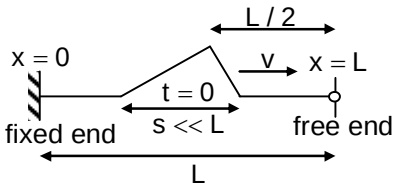
(D) Electric field will be zero outside the cable

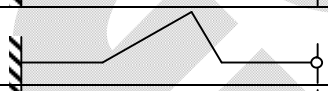
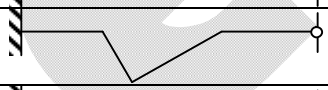
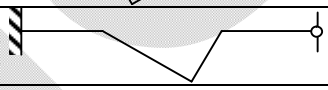
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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. A small pulse travelling with speed v in a string is shown at $t = 0$, moving towards free end. Select the shape of string from List II at moments shown in List I.

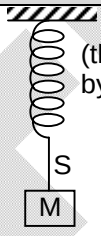
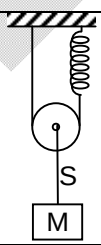


List - I		List - II	
(I)	$t = \frac{L}{v}$	(P)	
(II)	$t = \frac{2L}{v}$	(Q)	
(III)	$t = \frac{3L}{v}$	(R)	
(IV)	$t = \frac{4L}{v}$	(S)	
		(T)	

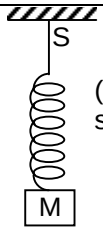
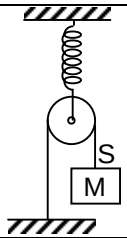
Codes:

	I	II	III	IV
(A)	Q	S	P	R
(B)	S	Q	R	P
(C)	P	S	R	Q
(D)	S	P	Q	R

8. In the following four situations, mass M of 1kg is kept in equilibrium. $k = 100$ N/m in all cases. What speed can be given to mass M vertically so that inextensible string S does not become slack in subsequent motion. Consider that pulley is ideal and string is mass less:

List - I		List - II	
(I)	 (the block is attached to spring by an inextensible thread)	(P)	$1\ ms^{-1}$
(II)		(Q)	$0.5\ ms^{-1}$

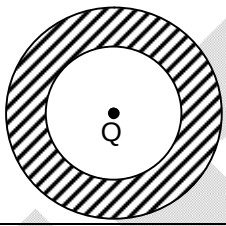
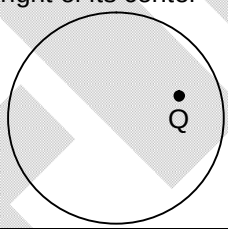
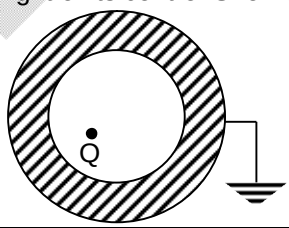
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(III)	 (1 kg block is attached to the spring)	(R)	0.25 ms^{-1}
(IV)		(S)	2 ms^{-1}
		(T)	3 ms^{-1}

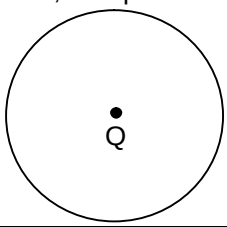
Codes:

	I	II	III	IV
(A)	PQT	T	RS	PQ
(B)	QS	PT	PST	PQ
(C)	PQR	R	PQR	PQRS
(D)	PQR	S	PQR	RS

9. Match List-I with List-II:

List - I		**List - II**	
(I)	Electrically neutral thick conducting spherical shell, with point charge at its center. 	(P)	Electric field everywhere inside the cavity due to charges induced on the inner surface of conductor is Zero.
(II)	Electrically neutral thin conducting spherical shell, with point charge to the right of its center 	(Q)	Electric field everywhere inside the cavity due to charges induced on the outer surfaces of conductor is zero
(III)	Electrically neutral thick conducting spherical shell, with point charge to the right of its centre. Shell is grounded. 	(R)	Electric potential at the centre of cavity due to charges induced on inner & outer surface of conductor is zero.

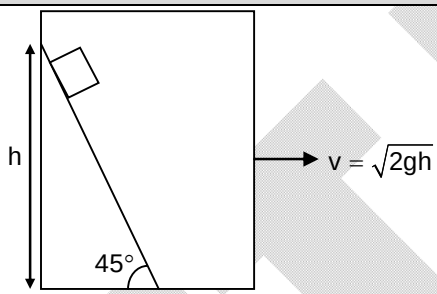
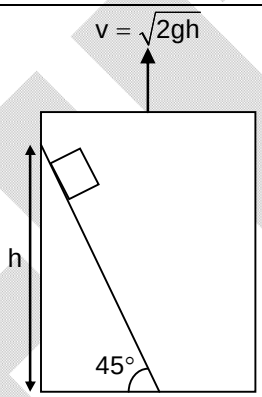
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(IV)	Electrically neutral thin conducting shell, with point charge at the centre. 	(S)	Electric potential everywhere inside the cavity due charges induced on the inner and outer surface conductor is zero.
		(T)	Net electric field outside the conductor is zero.

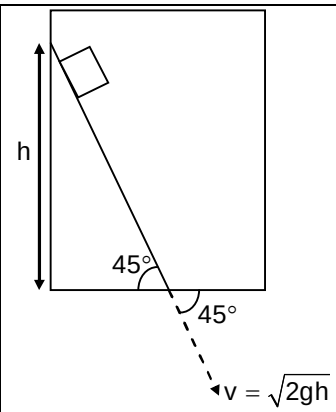
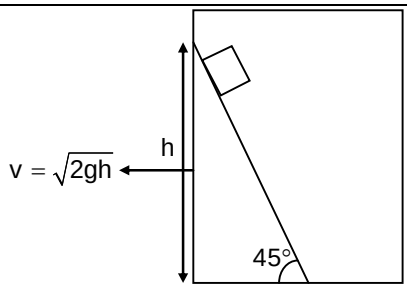
Codes:

	I	II	III	IV
(A)	PQR	PQ	RS	QRS
(B)	PQRS	QS	RST	PQR
(C)	QST	QRS	PQRS	PRS
(D)	PQ	QR	QT	PQRS

10. Figure shown four situations in which a small block of mass 'm' is released from rest (with respect to smooth fixed wedge inside a box) as shown in figure. List II shows work done by normal reaction with respect to an observer who is stationary with respect to ground till block reaches at the bottom of inclined wedge, match the appropriate list.

List - I		List - II	
(I)		(P)	Positive
(II)		(Q)	Negative

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(III)		(R)	Equal to mgh in magnitude
(IV)		(S)	Equal to zero
		(T)	Equal to $\frac{mgh}{2}$ in magnitude

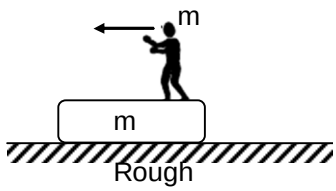
Codes:

	I	II	III	IV
(A)	P	T	Q	R
(B)	P	P	S	Q
(C)	S	Q	T	R
(D)	T	P	Q	R

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

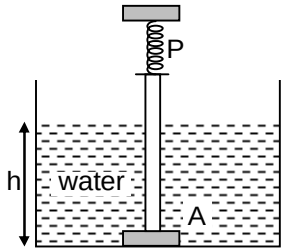
11. A block of mass m and length ℓ = 2 m is kept at rest on a rough horizontal ground of friction coefficient $\mu_k = 0.4$. A man of mass m is standing at the right end. Now the man starts walking towards left and reaches the left and within time t = 0.5 s. During this time, the displacement of the block (in m) is: (Assume the pressing force between the block and the ground remains constant and its value is same as it was initially. Also assume that the block slides during the entire time (t)):



12. Radius of a nucleus is given by the relation $R = R_0 A^{1/3}$ where $R_0 = 1.3 \times 10^{-15}$ m and A is mass number. For a nucleon inside a nucleus, de-Broglie wavelength is given by diameter of the nucleus. Average kinetic energy (in MeV) of a nucleon in the Te^{128} nucleus based on above information will be:

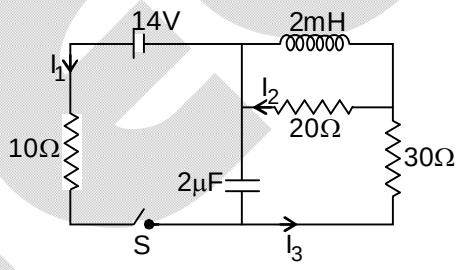
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13. In the figure shown there is a massless rigid rod at one end connected to a thin plate of mass $m = 1\,kg$ and area $A = 10\ cm^2$ which is covering a hole in a container filled with water of density $\rho = 1\,gm/cc$ to a height $h = 1\,m$. The other end of the rod is connected to a massless spring of stiffness $k = 40\ N/m$ on to which a block of same mass $m = 1\,kg$ is placed in equilibrium. Find the maximum compression from equilibrium position that can be given to the block so that after releasing it, liquid does not come out of container. $\left(g = 10\ m/s^2\right)$

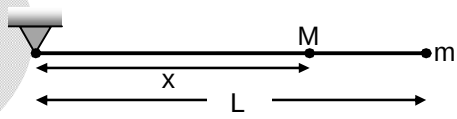


14. A cylinder-piston arrangement contains air under a pressure $P_1 = 10^4$ Pa . There is a soap bubble of radius r = 1 mm in the cylinder and its surface tension is T = 0.5 N/m. The soap bubble is enlarged to double its radius by pulling the piston out, temperature remaining constant. Determine the final pressure P_2 (in Pa) of the air.

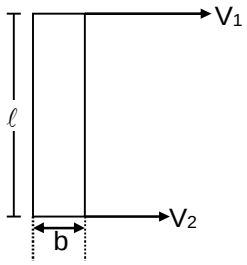
15. In the circuit shown, the switch is closed at $t = 0$, the currents I_1, I_2 & I_3 are as shown. Find the value of $I_1 + I_2 + I_3$ (in A).



16. A massless stick of length L = 2 m is hinged at one end and a mass m attached to its other end. The stick is free to rotate in vertical plane about a fixed horizontal axis passing through frictionless hinge. The stick is held in a horizontal position. At what distance x (in m) from the hinge should a second mass $M = m$ be attached to the stick, so that stick falls as fast as possible when released from rest?



17. A plank of length $l = 1\,m$ and width $d = 1\,mm$ is moving on a wet surface such that velocity of one and is $V_1 = 5$ m/s and velocity of other and is $V_2 = 2$ m/s as shown. Find viscous force on the rod of thickness of liquid layer below the plank is 0.1 mm and coefficient of viscosity is 0.1 Pa-s.



18. A well of depth $\frac{R}{2}$ is dug on the earth where R is radius of the earth. A particle is released from rest in the well. Time taken by the particle to reach the bottom of well is $\frac{\pi}{n}\sqrt{\frac{R}{g}}$, where g is acceleration due to gravity at the surface of earth. Find n.

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Chemistry

PART – II

Section – A (Maximum Marks: 24)

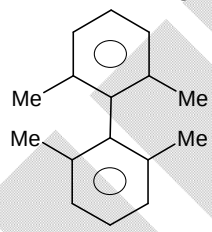
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

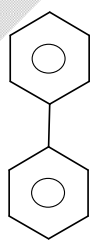
19. Which of the following is a correct statement ?
(A) Calamine is the ore of zinc.
(B) Proustite is the ore of silver.
(C) Cassiterite is the ore of tin.
(D) Diaspore is the ore of aluminium.

20. Which is/are correct statement regarding pH of NaA aq. solution?
(A) Approximate pH of 10^{-5} molar NaA aq. solution is 8.78 (If pKa of HA is 9, assume NaA is a completely soluble salt of NaOH and HA)
(B) Approximate pH of 10^{-5} molar NaA aq. solution is 9 (If pKa of HA is 9, assume NaA is a completely soluble salt of NaOH and HA)
(C) Approximate pH of 10^{-5} molar NaA aq. solution is 7 (If pKa of HA is 6, assume NaA is a completely soluble salt of NaOH and HA)
(D) Approximate pH of 10^{-5} molar NaA aq. solution is 7.5 (If pKa of HA is 6, assume NaA is a completely soluble salt of NaOH and HA)

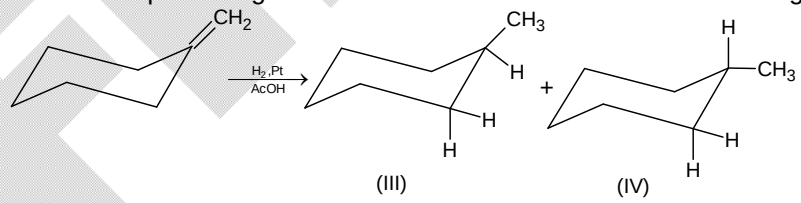
21. In ethane, the staggered conformation is more stable than eclipsed conformation. This is mainly due to:
(A) Steric hindrance between two H-atoms in eclipsed conformation
(B) The electrons in C–H bonds repels each other and this repulsion is maximum in eclipsed conformation
(C) Stabilizing interaction between C–H σ bonding orbital on one carbon atom and the C–H σ^* anti bonding orbital on the other carbon atom is maximum in staggered conformation
(D) All the three

22. Which of the following is/are correct?

(A) 

(I) is more reactive than  (II) towards EAS because of the extra methyl groups in compound (I)

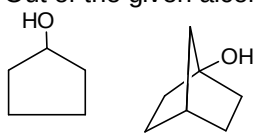
(B) The correct percentage of III and IV are 50% and 50% each in the given reaction.



(III) (IV)

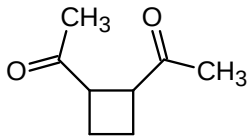
(C) The product of the reaction between $CH_3CH_2CH=CH_2$ and D_2 is optically inactive but resolvable.

(D) Out of the given alcohols, (V) is most reactive towards dehydration.



(V) (VI)

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23. 

$\xrightarrow[CF_3CO_3H]{1\text{ eq.}}$ A $\xrightarrow{HCN}$ B $\xrightarrow[(ii)H^+]{(i)KOH}$ C + D

Which is/are correctly matched

	Compound	Qualitative Test
(A)	C and D both	Br_2 water decolourization
(B)	C and D both	Give effervescence with baking soda solution
(C)	C and D both	Red coloration to ceric ammonium nitrate solution
(D)	C and D both	Upon dry heating loose CO_2.

24. Given below are four reactions. The reactions in which sum of co-efficient of reactants after balancing either 3 or 4 are:

(A) $Cl_2 + SeO_3^{2-} + H_2O \longrightarrow SeO_4^{2-} + Cl^- + H^+$ (B) $FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

(C) $Cl_2 + IO_3^- + OH^- \longrightarrow Cl^- + IO_4^- + H_2O$ (D) $Br_2 + OH^- \longrightarrow BrO^- + Br^- + H_2O$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the reaction mechanisms mentioned in List – I with their characteristics mentioned in List – II.

List - I		List - II	
(I)	S_N1	(P)	The reactivity order of alkyl halides is $1^o > 2^o > 3^o$
(II)	S_N2	(Q)	The reactivity order of alky halides is $3^o > 2^o > 1^o$
(III)	E_1	(R)	Reaction intermediate(s) is/are formed
(IV)	E_2	(S)	Rate decreases with increase in polarity of solvent
		(T)	Reaction intermediate is carbene

Codes:

	I	II	III	IV
(A)	ST	RS	QP	QS
(B)	QP	RT	QS	QR
(C)	SR	RS	QP	QR
(D)	QR	PS	QR	QS

26. In the reactions in List-I, n-factor of underlined molecules are given in List-II. Match List-I with List-II:

List – I (underlined molecules)		List – II (n-factor)	
(I)	$\underline{K_4[Fe(CN)_6]} \longrightarrow Fe^{3+} + CO_2 + NO_3^-$	(P)	61
(II)	$\underline{FeS_2} \longrightarrow Fe^{3+} + SO_2$	(Q)	11
(III)	$\underline{Br_2} + 2NaOH \longrightarrow NaOBr + NaBr + H_2O$	(R)	1
(IV)	$\underline{As_2S_3} \longrightarrow As^{+5} + SO_4^{2-}$	(S)	5/3
		(T)	28

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Codes:

	I	II	III	IV
(A)	P	Q	R	S
(B)	P	Q	S	T
(C)	Q	P	S	R
(D)	S	R	T	P

27. Match List-I with List-II:

List – I		List – II	
(I)	In 1 unit cell of hcp packing the number of octahedral voids are present	(P)	6
(II)	In 1 unit cell of hcp packing the number of tetrahedral voids are present	(Q)	12
(III)	In bcc tetrahedral voids are present in between two body centre atoms	(R)	Tetrahedral voids are not possible in square arrange, i.e., BCC
(IV)	Perfect ionic crystal do not conduct electricity	(S)	Entropy is zero for perfect crystal at zero Kelvin
		(T)	at $\frac{a\sqrt{3}}{6}$ height (a = edge length)

Codes:

	I	II	III	IV
(A)	S	Q	P	T
(B)	R	S	P	Q
(C)	P	Q	R	S
(D)	P	R	T	S

28. Match List-I (Characteristics of reaction) with List-II (Order of reaction):

List - I		List - II	
(I)	Rate of reaction is independent of concentration of reactant	(P)	0
(II)	Reaction rate doubles on increasing the reactant concentration four times	(Q)	1
(III)	Half life period is inversely proportional to initial concentration of reactant	(R)	2
(IV)	Half life period is independent of concentration	(S)	1/2
		(T)	–1/2

Codes:

	I	II	III	IV
(A)	T	R	P	Q
(B)	P	S	R	Q
(C)	S	T	R	P
(D)	Q	S	P	R

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. The total possible stereoisomer for the molecule $[Ma_2bcde]^{m\pm}$ is x, then $\frac{x}{4}$ is .....

(where a, b, c, d, e are simple monodentate ligand having no chiral centre)

30. If x is the maximum number of equal P – O bonds in $P_2O_7^{4-}$ ion. Then $\frac{x}{4} = ?$

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31. A very small amount of a non-volatile solute (non-associative, non-dissociative) is dissolved in 100 cm^3 of a solvent. At room temperature, vapour pressure of this solution is 98.7 mm of Hg while that of pure solvent is 100 mm of Hg. If the freezing temperature of this solution is 0.72 K lower than that of pure solvent, what is the value of cryoscopic constant of solvent (in K Kg/mol)? Given : Molar mass of solvent = 78 g/mol.

32. Formaldehyde reacts with ammonia to produce a compound which is used to treat urinary infections. X number of tertiary nitrogen atoms are present in one molecule of the product formed. Then value of $\frac{X}{3}$ is

33. How many degenerate atomic orbital(s) is/are present in the third orbit of hydrogen atom?

34. If volume occupied by CO_2 molecules is negligible then calculate pressure exerted by one mole of CO_2 gas at 300K. a = 3.592 atm $litre^2 mol^{-2}$.

35. A sample of Br_2 containing some inert impurity undergo complete reaction with 60ml of hot & conc. NaOH solution. The resulting solution is treated with excess of $AgNO_3$ solution where all Br^- produce in the reaction is precipitated out in the form of AgBr. The precipitate is isolated & treated with $\frac{2}{3}$ M $KMnO_4$ solution in neutral medium & required 50 ml of $KMnO_4$ to convert all Br^- into Br_2. Calculate the molarity of the NaOH solution used.

36. The velocity of a reaction is tripled for every 10^oC rise in temperature. If temperature is raised by 120^oC then the velocity increases by x times. $\frac{x}{729}$ is equals to:

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Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. Let $A(z_1)$ and $B(z_2)$ are two complex numbers in the complex plane and 'O' is origin. If $\frac{z_1}{z_2}+\frac{\bar{z}_1}{\bar{z}_2}=2$, then

(A) $\angle B=\frac{\pi}{3}$ (B) $\angle B=\frac{\pi}{2}$

(C) $\angle AOB=\frac{\pi}{3}$; if $|z_1|=2|z_2|$ (D) $\angle AOB=\frac{\pi}{6}$, if $|z_1|=2|z_2|$

38. Let $a,b,\in N$, $a\neq b$ and the quadratic equations $(a-1)x^2-(a^2+2)x+a^2+2a=0$, $(b-1)x^2-(b^2+2)x+b^2+2b=0$ have a common root, then

(A) a = 3 (B) a = 2

(C) b = 2 (D) a = 4

39. A rectangular Hyperbola has one of the asymptotes $x-y+2=0$ and tangent at $P(0,1)$ is x = 0, thus,

(A) Length of latus rectum is 2

(B) Center of Hyperbola is (-1, 1)

(C) Equation of other asymptote is $x+y-1=0$

(D) Length of latus rectum is 4

40. Consider a differentiable function $f:R-R$ such that $f(x+y)=3^x.f(y)+9^y f(x)\forall\ x,y\in R$ also $f'(0)=\ln 3$, then

(A) $f(1)=6$ (B) $f\left(\log_3^2\right)$

(C) $f(2)=72$ (D) $f(1)=9$

41. Let $f(x)=x^{13}+2x^{12}+3x^{11}+....+13x+14$ and $\alpha=e^{i\frac{2\pi}{15}}$, if $N=f(\alpha)f(\alpha^2)f(\alpha^3).....f(\alpha^{14})$ then

(A) Number of divisors of N is 196

(B) Number of divisors of N is 956

(C) Number of divisors of N which are perfect squares is 49

(D) Number of divisors of N which are perfect cube is 16

42. A variable circle 'C' touches the lies y = 0 and passes through the point (0, 1) let the locus of the center of the circle 'C' is P, then

(A) Equation of P is $2y=1+x^2$

(B) Area bounded by curve P and line $x+y=2$ is $\frac{16}{3}$

(C) Area bounded by curve P and line x + y = 4 is $\frac{16}{3}$

(D) Equation of P is $2y=2+x^2$

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Match List-I with List-II:

List - I		List - II	
(I)	Number of points where the function $f(x)=\max\{\lvert\tan\pi x\rvert,\ \cos\pi x\}$ is non differentiable in (-1, 1)	(P)	9
(II)	The sum factors of $\lfloor 7$, which are the form of 3n + 1 and odd ($n \in W$)	(Q)	4
(III)	If $ax^2+bx+c=0$ and $2x^2+3x+4=0$ have a common root and a, b, $c \in N$, then least value of a + b + c	(R)	1
(IV)	Tangents are drawn from $(-2,0)$ to $y^2=8x$, if radius of circle is 'r' which touches tangents drawn from (-2, 0) and corresponding chord of contact, the value of $[r]$ is (where [.] denotes greatest integer value)	(S)	9
		(T)	8

Codes:

	I	II	III	IV
(A)	T	Q	P	QR
(B)	Q	T	P	QR
(C)	P	Q	T	S
(D)	RS	P	Q	T

44. If P is symmetric and Q is skew symmetric matrix and P + Q is non singular matrix, where order of P and Q are equal of $A=(P+Q)^{-1}\cdot(P-Q)$ then

List - I		List - II	
(I)	$\text{Det}\left(A^T(P+Q)A\right)$ is	(P)	$\frac{\text{Det}(P-Q)}{\text{Det}(P+Q)}$
(II)	$\text{Det}(P+Q)A(P-Q)^T$	(Q)	$\frac{\text{Det}(P+Q)}{\text{Det}(P-Q)}$
(III)	$\text{Det}(A)$	(R)	$\text{Det}(P-Q)$
(IV)	$\text{Det}\left(A^{-1}\right)$	(S)	$\text{Det}\left(P^2-Q^2\right)$
		(T)	Det(P+ Q)

Codes:

	I	II	III	IV
(A)	T	S	P	Q
(B)	S	R	P	Q
(C)	P	Q	R	S
(D)	S	T	P	R

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45. Match List-I with List-II:

List - I		List - II	
(I)	The values of m for which the roots x_1 and x_2 of the quadratic equation $x^2 - 2mx_3 + m = 0$ satisfy the condition $x_1^3 + x_2^3 = x_1^2 + x_2^2$ is	(P)	4
(II)	If the system of equation $x + 2y + 3z = 4$, $x + py + 2z = 3$, $x + 4y + \mu z = 3$ has an infinite number of solution, then $p + \mu$ is	(Q)	12
(III)	The number of solution of the matrix equation $x^2 = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$ is	(R)	6
(IV)	If A, B and C are $n \times n$ matrices and det(A) = 2, det(B) = 3 and det(C) = 5, then the value of $5\det\left(A^2BC^{-1}\right)$ is equal to	(S)	1/4
		(T)	0

Codes:

	I	II	III	IV
(A)	P	Q	R	T
(B)	ST	R	P	Q
(C)	R	P	T	S
(D)	Q	T	RT	S

46. Match List-I with List-II:

List - I		List - II	
(I)	Let $\vec{\pi} = \left(\vec{a} \times \vec{b}\right)\sin x + \left(\vec{b} + \vec{c}\right)\cos y + \vec{c} \times \vec{a}$ where $\vec{a}, \vec{b}, \vec{c}$ are non zero non coplanar vectors. If $\vec{\pi}$ is orthogonal to $3\vec{a} + 5\vec{b} + 2\vec{c}$, the value of $\sin^2 x + \cos^2 y + 15$ is	(P)	11
(II)	The equation of circle having the lines $x^2 + 2xy + 3x + 6y = 0$ as its normal, having size just sufficient to contains the circle $x(x-4) + y(y-3) = 0$ is $x^2 + y^2 + 6x - 3y - k = 0$ then k is	(Q)	17
(III)	The maximum area of the rectangle where base is on x axis and two of its vertices lines on the curve $y = e^{-x^2}$ is $e^{\left(\sqrt{k} - \frac{1}{k}\right)}$ then value of 22k is,	(R)	44
(IV)	A regular decagon have perimeter p and area of decagon is $\frac{p^2}{\mu - 1}\cot\frac{\pi}{10}$, then μ takes value	(S)	45
		(T)	38

Codes:

	I	II	III	IV
(A)	P	R	S	P
(B)	R	Q	S	T
(C)	Q	S	R	Q
(D)	S	T	Q	P

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Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. The number of complex numbers satisfying the equation $z^2 = z + |z|^2 + 2$ is

48. If α,β,γ are the roots of the equation $x^3 + x^2 - 2x + 1 = 0$ then find the value of

$$\begin{vmatrix} \gamma^2 & \beta^2 & 2\beta\gamma - \alpha^2 \\ 2\alpha\gamma - \beta^2 & \alpha^2 & \gamma^2 \\ \alpha^2 & 2\alpha\beta - \gamma^2 & \beta^2 \end{vmatrix}$$

49. There are 'n' natural number in the set. If one number is missing from the set, and the AM of remaining n – 1 number is $\frac{602}{17}$, then find the missing number.

50. Let a_1,a_2,a_3,a_4,a_5 are five terms in geometric series satisfying the condition $0 < a_1 < a_2 < a_3 < a_5 < 100$ where each term is an integer then number of sets $\{a_1,a_2,a_3,a_4,a_5\}$ is.

51. The equation $x^2 + bx + c = 0$ has two distinct roots. If 2 is subtracted from each root the rest are the reciprocal of the original roots then $b^2 + c^2$ is.

52. The coefficient of x^{1274} in the expansion of $(x+1)(x-2)^2(x-3)^3(x-4)^4 .....(x-49)^{49}(x-50)^{50}$ is k, then value of |k| is.

53. Two boxes containing 20 balls each and each ball is either black or white. The total number of black balls is different from the total number of white balls. One ball is drawn at random from each box. If the probability that both balls are white is 0.21, then the probability that both the balls drawn are black, is P. Then find the value of 50P.

54. A complex number z with Im(z) = 4 and a positive integer 'n' such that $\frac{z}{z+n} = 4i$, then find the value of n.

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